

Results: LightGBM vs. Transformer

Early-warning degradation prediction, held-out EVAL block, machine-1-1 (SMD)

Both models predict P(degradation within the next 30 minutes) from a 20-minute KPI window (Component 1 of the trustworthy-RCA framework). LightGBM is the auditable tabular baseline (rolling mean/std/last features); the Transformer is a 2-layer self-attention encoder over the raw window. Both are evaluated on the same held-out EVAL block, never used for fitting. Source: notebooks/02_prediction_baselines.ipynb, seeds [42, 123, 2026, 3407, 999].

Summary (mean $\pm$ std across 5 seeds)

Model	AUROC	AUPRC
LightGBM	0.898 (identical across all 5 seeds)	0.825 (identical across all 5 seeds)
Transformer	0.926 $\pm$ 0.010	0.880 $\pm$ 0.013

Per-seed detail

Seed	LightGBM AUROC	LightGBM AUPRC	Transformer AUROC	Transformer AUPRC
42	0.898	0.825	0.909	0.862
123	0.898	0.825	0.931	0.875
2026	0.898	0.825	0.933	0.882
3407	0.898	0.825	0.932	0.898
999	0.898	0.825	0.926	0.881

Statistical comparison

Paired Wilcoxon signed-rank test across the 5 matched seeds, on AUPRC: mean 0.880 (Transformer) vs. 0.825 (LightGBM), $p = 0.0625$. Not significant at $\alpha = 0.05$.

Caveats

- LightGBM's variance is exactly 0.000 across seeds — not an error. With bagging_fraction/feature_fraction at their default 1.0, there is no row/column subsampling for random_state to actually randomize, so this configuration is fully deterministic regardless of seed.
- The Transformer scores higher in 5 of 5 seeds, yet the test still cannot reach $p < 0.05$. This is not a contradiction: at $n = 5$ paired samples, $p = 0.0625$ is the smallest two-sided Wilcoxon p-value achievable when every pair agrees in direction — $n = 5$ is structurally incapable of reaching 0.05 no matter how consistent the effect is. The honest conclusion is that the Transformer is directionally ahead in every seed tried, but this experiment cannot statistically confirm it as the better model; more seeds (and more machines) are required before that claim is supported rather than observed as a trend.